

Trainings ▾

Preparatory Steps

Automotive with CM875, CM870 and CM570

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

- Disconnect the batteries. Refer to Procedure 013-009 in Section 13 (</qs3/pubsys2/xml/en/procedures/99/99-013-009.html>).
- Drain the coolant. Refer to Procedure 008-018 in Section 8 (</qs3/pubsys2/xml/en/procedures/35/35-008-018-tr.html>).

Automotive with CM876

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Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing,

remove the negative (-) battery cable first and attach the negative (-) battery cable last.

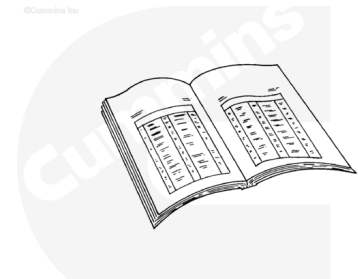
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⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

- Disconnect the batteries. Refer to Procedure 013-009 in Section 13 (</qs3/pubsys2/xml/en/procedures/99/99-013-009.html>).
- Remove the turbocharger oil supply line. Refer to Procedure 010-045 in Section 10 (</qs3/pubsys2/xml/en/procedures/35/35-010-045-tr.html>).
- Remove the oil drain lines from the turbocharger. Refer to Procedure 010-046 in Section 10 (</qs3/pubsys2/xml/en/procedures/35/35-010-046-tr.html>).
- Drain the coolant. Refer to Procedure 008-018 in Section 8 (</qs3/pubsys2/xml/en/procedures/35/35-008-018-tr.html>).



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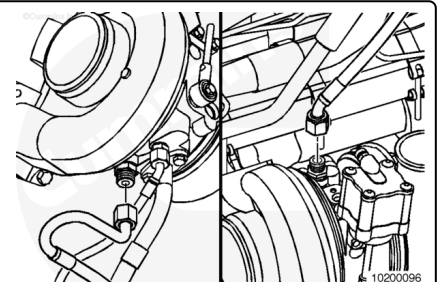
Remove

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Use a second wrench to hold the coolant line coupling while loosening the line.

Remove the turbocharger coolant supply line (bottom) from the exhaust gas recirculation (EGR) coolant supply tube.

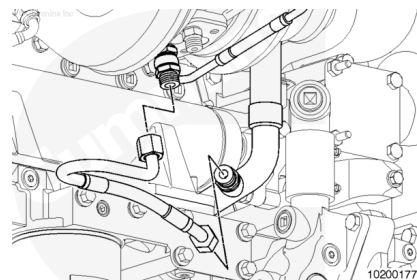
Remove the turbocharger return (top) line from the water manifold and the turbocharger.



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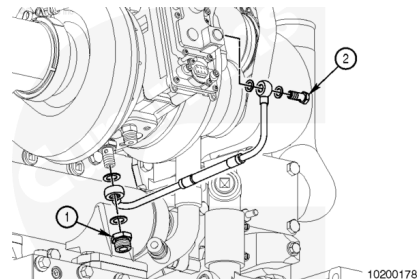
Note : Use a second wrench to hold the coolant line coupling while loosening the line.

Remove the turbocharger coolant supply line from the bottom of the turbocharger and the EGR cooler coolant supply tube.



Remove the turbocharger actuator coolant supply line from the bottom of the turbocharger bearing housing by removing the retaining nut, the banjo bolt, and washers from the actuator.

Remove the adapter.

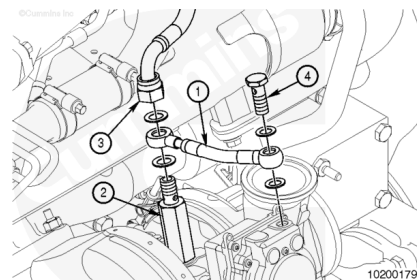


Note : Use a second wrench to hold the coolant line coupling while loosening the line.

Remove the turbocharger coolant return line from the top of the turbocharger and the water manifold housing.

Remove the turbocharger actuator coolant return line (1) from the turbocharger coolant-out fitting adapter (2) by loosening the retaining nut (3) and the banjo bolt and washers (4).

Remove the coolant supply and return fittings from the turbocharger bearing housing if replacing it with a new component. (Service turbochargers are **not** supplied with these fittings.)



Clean and Inspect for Reuse

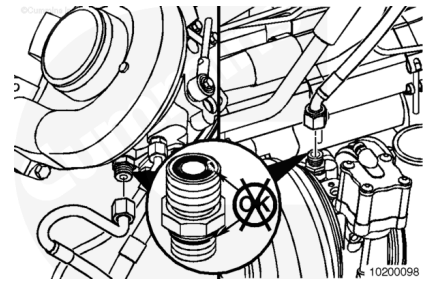
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⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



Clean the turbocharger coolant lines with solvent.

Dry with compressed air.

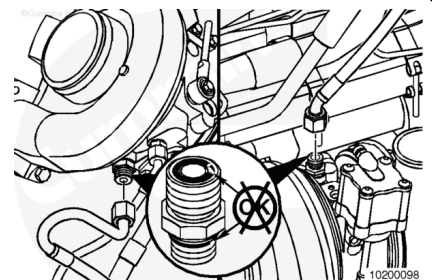
Check the o-rings for cuts or deformation. Replace if damage is found.

Check the turbocharger coolant lines for cracks or other damage. Replace if damage is found.

Automotive with CM876

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⚠ WARNING ⚠

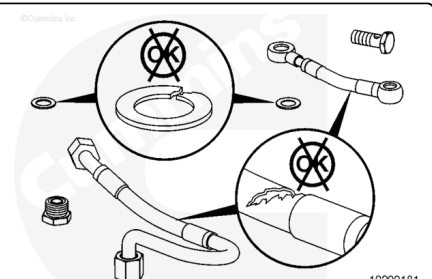
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Clean the turbocharger coolant lines with solvent.

Dry with compressed air.

Check the o-rings and banjo washers for cuts or deformation. Replace if damage is found.

Check the turbocharger coolant lines for cracks or any other damage. Replace if damage is found.



Check the face of the coolant fittings on the engine and on the turbocharger to make sure the o-ring is in place and is **not** damaged. A damaged or missing o-ring will result in a coolant leak.

Install

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⚠ CAUTION ⚠

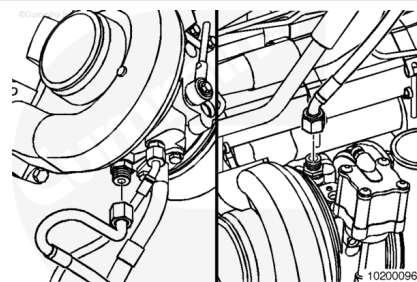
Do not bend or kink hoses while tightening. Doing so may damage the line.

⚠ CAUTION ⚠

Do not use thread sealant on fittings. Doing so can possibility damage internal components.

⚠ CAUTION ⚠

Check the face of the coolant fitting on the engine and on the turbocharger to make sure o-ring is in place and is not damaged. A damaged or missing o-ring will result in a coolant leak.



Use a second wrench to hold the coolant line coupling while tightening the line.

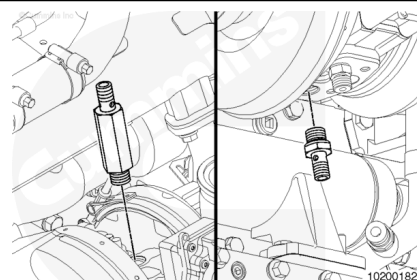
Install the turbocharger coolant supply line to the EGR supply and the turbocharger, and the return line to the water manifold and turbocharger.

Torque Value: 46 n•m [34 ft-lb]

Automotive with CM876

⚠ CAUTION ⚠

Do not use petroleum products on the banjo sealing washers. The sealing material will swell and cause the connection to leak.



Install the coolant inlet and outlet fitting adapters into the turbocharger bearing housing.

Torque Value: 54 n•m [40 ft-lb]

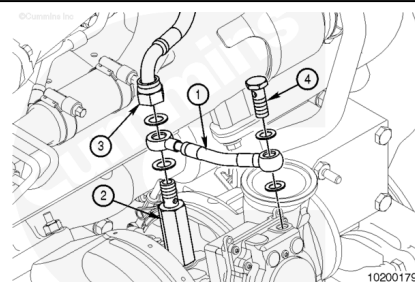
Lubricate the banjo sealing washer with clean engine coolant, soapy water, or vegetable oil.



Install the banjo washers on each side of both coolant return tube ports.

Install the turbocharger actuator coolant return line (1) to the turbocharger coolant outlet fitting (2) by installing the retaining nut (3). Do **not** torque.

Install the banjo bolt. Tighten the banjo bolt and washers (4).



Torque Value: 24 n•m [18 ft-lb]

Tighten the turbocharger coolant outlet retaining nut (3).

Torque Value: 54 n•m [40 ft-lb]

Install the turbocharger coolant return line on top of the turbocharger and the water manifold housing.

Torque Value: 46 n•m [34 ft-lb]

⚠ CAUTION ⚠

Do not use petroleum products on the banjo sealing washers. The sealing material will swell and cause the connection to leak.

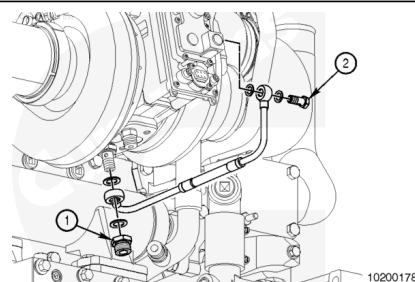


Lubricate the banjo sealing washer with clean engine coolant, soapy water, or vegetable oil.

Install the banjo washers on each side of both coolant supply tube ports.

Install the turbocharger coolant supply line to the bottom of the turbocharger bearing housing by installing the retaining nut (1). Do **not** torque.

Install the turbocharger actuator coolant supply banjo bolt (2).



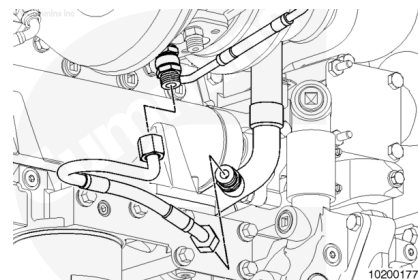
Torque Value: 15 n•m [132 in-lb]

Tighten the turbocharger coolant supply retaining nut (1).

Torque Value: 46 n•m [34 ft-lb]

Install the coolant supply line.

Torque Value: 46 n•m [34 ft-lb]

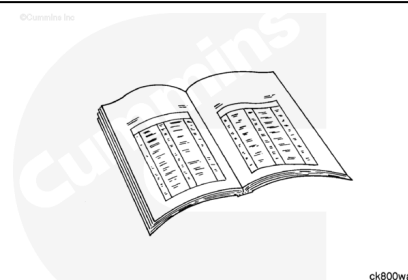


Finishing Steps

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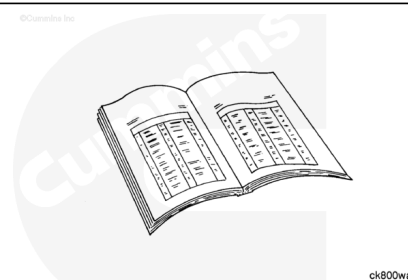
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- Fill the cooling system. Refer to Procedure 008-018 in Section 8 (/qs3/pubsys2/xml/en/procedures/35/35-008-018-tr.html).
- Connect the batteries. Refer to Procedure 013-009 in Section 13 (/qs3/pubsys2/xml/en/procedures/99/99-013-009.html).
- Operate the engine and check for leaks.

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Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



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- Install the turbocharger oil supply line. Refer to Procedure 010-045 in Section 10 (</qs3/pubsys2/xml/en/procedures/35/35-010-045-tr.html>).
- Install the oil drain lines to the turbocharger. Refer to Procedure 010-046 in Section 10 (</qs3/pubsys2/xml/en/procedures/35/35-010-046-tr.html>).
- Fill the coolant. Refer to Procedure 008-018 in Section 8 (</qs3/pubsys2/xml/en/procedures/35/35-008-018-tr.html>).
- Connect the batteries. Refer to Procedure 013-009 in Section 13 (</qs3/pubsys2/xml/en/procedures/99/99-013-009.html>).
- Operate engine and check for leaks.